DSA03-Natural Language Processing

List of Lab Program

8. Implement a simple stochastic part-of-speech tagging algorithm using a basic probabilistic model to assign POS tags using python.

Code:

```
import random

# Probabilistic model: word and its possible POS tags
pos_model = {
    "I": [("PRP", 1.0)],
    "love": [("VBP", 0.8), ("NN", 0.2)],
    "Python": [("NNP", 0.9), ("NN", 0.1)],
    "programming": [("NN", 0.7), ("VBG", 0.3)]
}
text = "I love Python programming"
words = text.split()
print("Word\t\tPOS Tag")
print("-" * 25)
for word in words:
    if word in pos_model:
        tags, probabilities = zip(*pos_model[word])
        tag = random.choices(tags, weights=probabilities)[0]
    else:
        tag = "UNKNOWN"
    print(f"{word:<16}{tag}")
```

Output:

